THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 8: **Introduction to Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\dfrac{\sin^2 40^\circ + \sin^2 50^\circ}{\cos^2 12^\circ + \cos^2 78^\circ} =$ 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** $\frac{1}{2}$

2. The value of $\tan 15^\circ \tan 35^\circ \tan 55^\circ \tan 75^\circ$ is: 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** $\tan 45^\circ + 1$

3. If $\sin\theta = \frac{15}{17}$ then $\sin^2\theta + \cos^2\theta + \tan^2\theta =$ 1

(a) $\frac{289}{64}$ **(b)** $\frac{64}{289}$ **(c)** $\frac{225}{289}$ **(d)** $\frac{225}{64}$

4. If $\tan\theta = \frac{15}{8}$, then $\dfrac{1-\tan^2\theta}{1+\tan^2\theta} =$ 1

(a) $-\frac{161}{289}$ **(b)** $\frac{161}{289}$ **(c)** $\frac{8}{15}$ **(d)** $\frac{240}{289}$

5. If $\sin\theta = \frac{15}{17}$, the value of $2\sin\theta\cos\theta$ is: 1

(a) $\frac{120}{289}$ **(b)** $\frac{30}{17}$ **(c)** $\frac{16}{17}$ **(d)** $\frac{240}{289}$

6. If $\sec\theta + \tan\theta = 4$, then $\sec\theta - \tan\theta =$ 1

(a) 1 **(b)** $\frac{1}{4}$ **(c)** 2 **(d)** 4

7. $\dfrac{\sin^2 58^\circ + \sin^2 32^\circ}{\cos^2 20^\circ + \cos^2 70^\circ} =$ 1

(a) 2 **(b)** $\frac{1}{2}$ **(c)** 0 **(d)** 1

8. The value of $\tan 25^\circ \tan 42^\circ \tan 48^\circ \tan 65^\circ$ is: 1

(a) 2 **(b)** 0 **(c)** $\tan 45^\circ + 1$ **(d)** 1

9. **Assertion (A):** $\sin\theta$ can be equal to 1.5 for some acute angle θ. 1

Reason (R): In a right triangle the hypotenuse is the longest side, so $\sin\theta = \dfrac{\text{perpendicular}}{\text{hypotenuse}} \le 1$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $\sin 30^\circ = \cos 60^\circ$. 1
Reason (R): $\tan 60^\circ = \sqrt{3}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $\cot\theta = \frac{9}{40}$, find $\sin\theta$ and $\sec\theta$. 2

12. If $x = a\sec\theta$ and $y = b\tan\theta$ with $a = 3$, $b = 5$, express $\frac{x^2}{9} - \frac{y^2}{25}$. 2

13. If $\sin\theta = \frac{15}{17}$, find the value of $\cos^2\theta - \sin^2\theta$. 2

14. If $\cot\theta = \frac{3}{4}$, find $\sin\theta$ and $\sec\theta$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that: $\dfrac{1-\tan^2 A}{1+\tan^2 A} = \cos^2 A - \sin^2 A$. 3

16. If $\sin\theta + \cos\theta = \frac{17}{13}$, find $\sin\theta\cos\theta$. 3

17. If $\tan\theta + \cot\theta = 5$, find $\tan^2\theta + \cot^2\theta$. 3

OR

Prove that: $(1+\tan^2 A)\sin^2 A = \tan^2 A$. 3

18. If $\sin\theta + \cos\theta = \frac{5}{4}$, find $\sin\theta\cos\theta$. 3

OR

If $\tan\theta + \cot\theta = 6$, find $\tan^2\theta + \cot^2\theta$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Riya is practising standard-angle values for her unit test. 5
(i) Find $\sin 30^\circ + \cos 60^\circ$.
(ii) Find $\tan 60^\circ \times \tan 30^\circ$.
(iii) Find $\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ$.
(iv) If $\sin(A+B) = 1$ and $\cos(A-B) = \dfrac{\sqrt{3}}{2}$, where $0^\circ < A+B \le 90^\circ$ and $A > B$, find A and B.

20. Prove that: $\dfrac{\sec\theta + \tan\theta - 1}{\tan\theta - \sec\theta + 1} = \dfrac{1+\sin\theta}{\cos\theta}$. 5

OR

If $\tan A = \dfrac{8}{6}$, evaluate $\dfrac{2\sin A + 3\cos A}{8\sin A + 1\cos A}$. 5

END OF QUESTION PAPER — Total: 40 Marks